

On the first bifurcation curve of the Hénon map

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Introduction

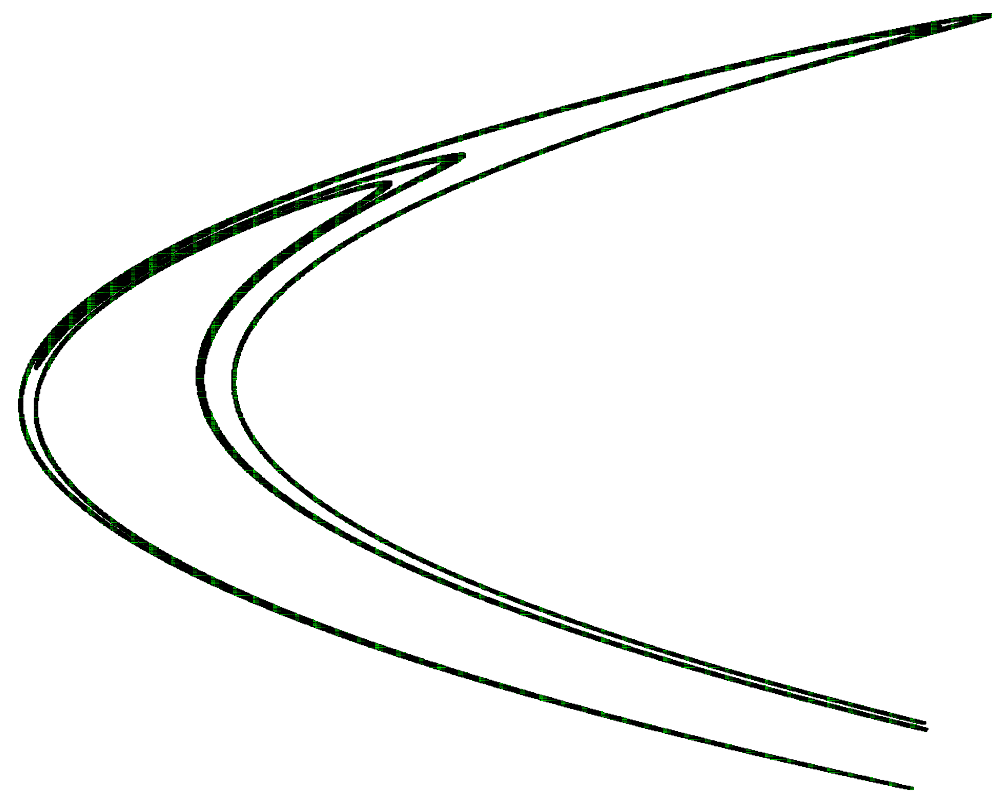
Hénon Map

The *Hénon family* on $\mathbb{R}^2$:

$$H_{a,b}(x, y) = (x^2 - a - by, x)$$



Michel Hénon
1931-2013



strange attractor for $(a, b) = (1.4, -0.3)$.

Fixed Points

For

$$a \geq -\frac{(1+b)^2}{4},$$

Hénon map has two fixed points

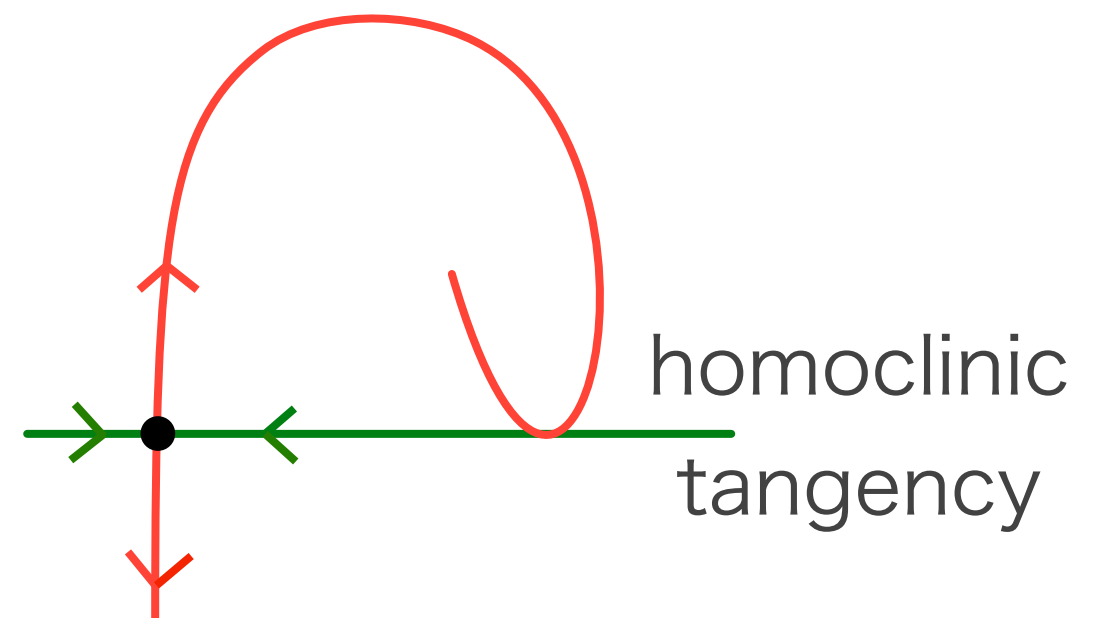
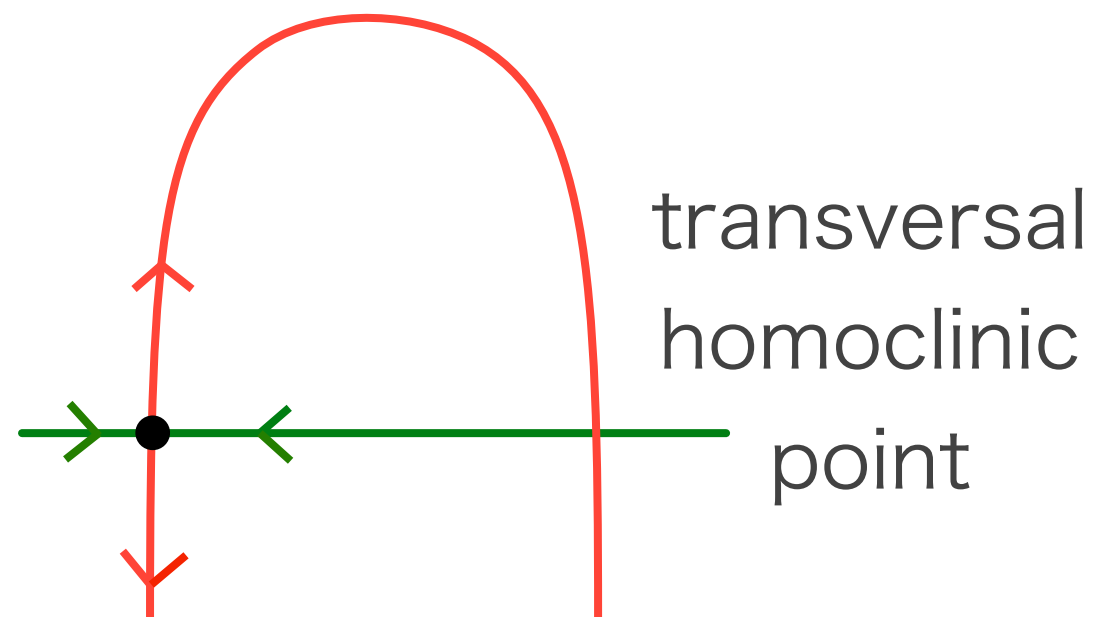
$$P = \left(\frac{b+1 + \sqrt{(b+1)^2 + 4a}}{2}, \frac{b+1 + \sqrt{(b+1)^2 + 4a}}{2} \right)$$

and

$$Q = \left(\frac{b+1 - \sqrt{(b+1)^2 + 4a}}{2}, \frac{b+1 - \sqrt{(b+1)^2 + 4a}}{2} \right)$$

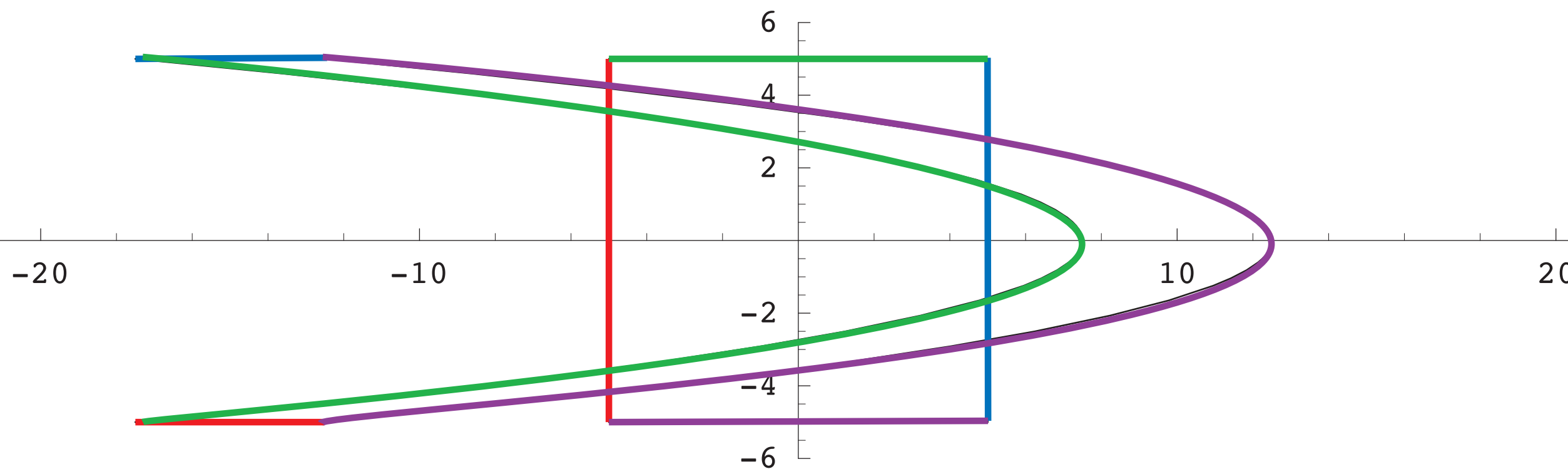
Homoclinic Points

Stable and unstable manifolds may have intersections.

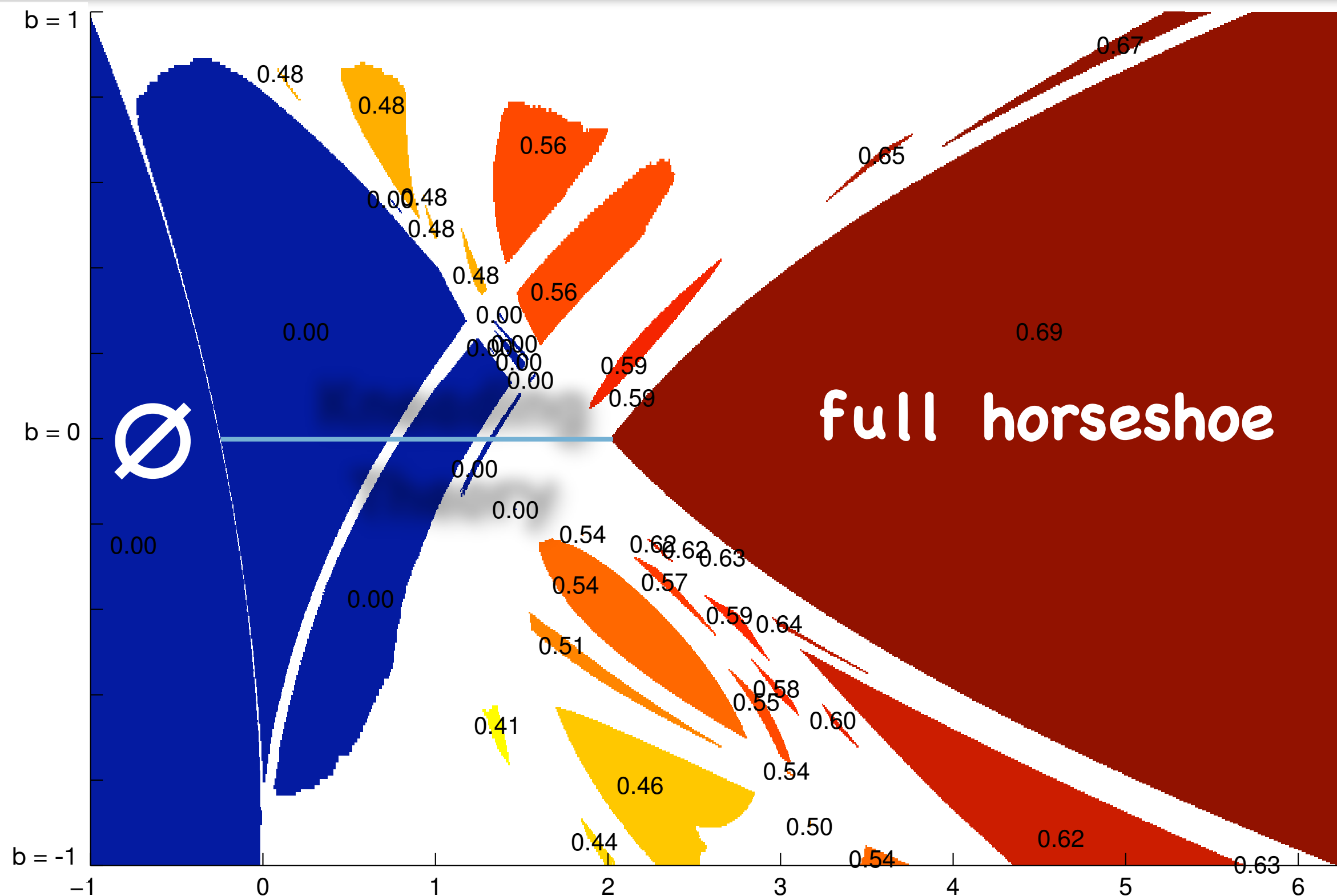


Horseshoe in the Hénon

[Devaney and Nitecki 1979] For any fixed b , if a is sufficiently large, then the non-wandering set of the Hénon map is uniformly hyperbolic full horseshoe.



Creation of horseshoe

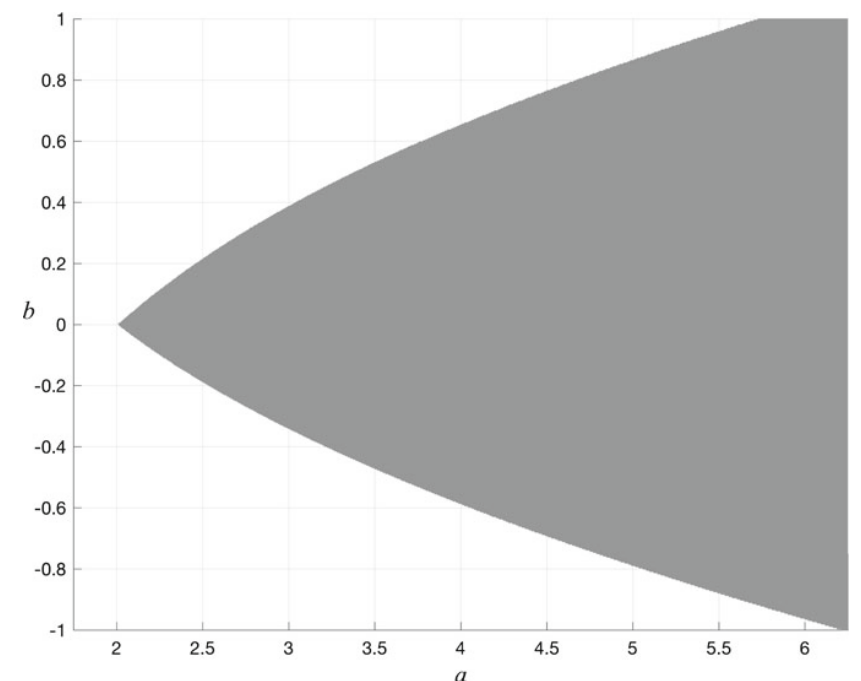


Hyperbolic plateaus of the real Hénon map $(x, y) \mapsto (a - x^2 + by, x)$
and rigorous lower bounds of the topological entropy

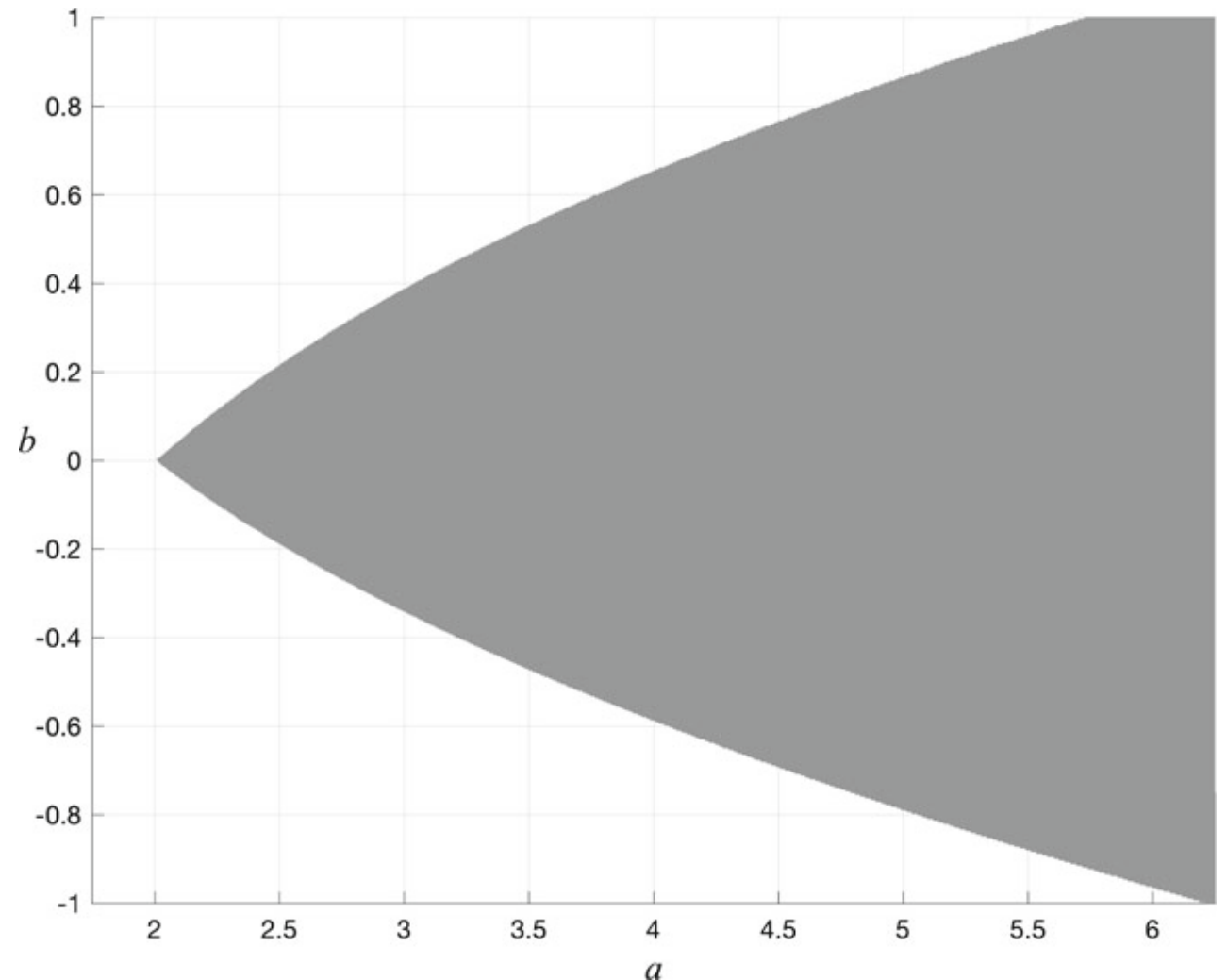
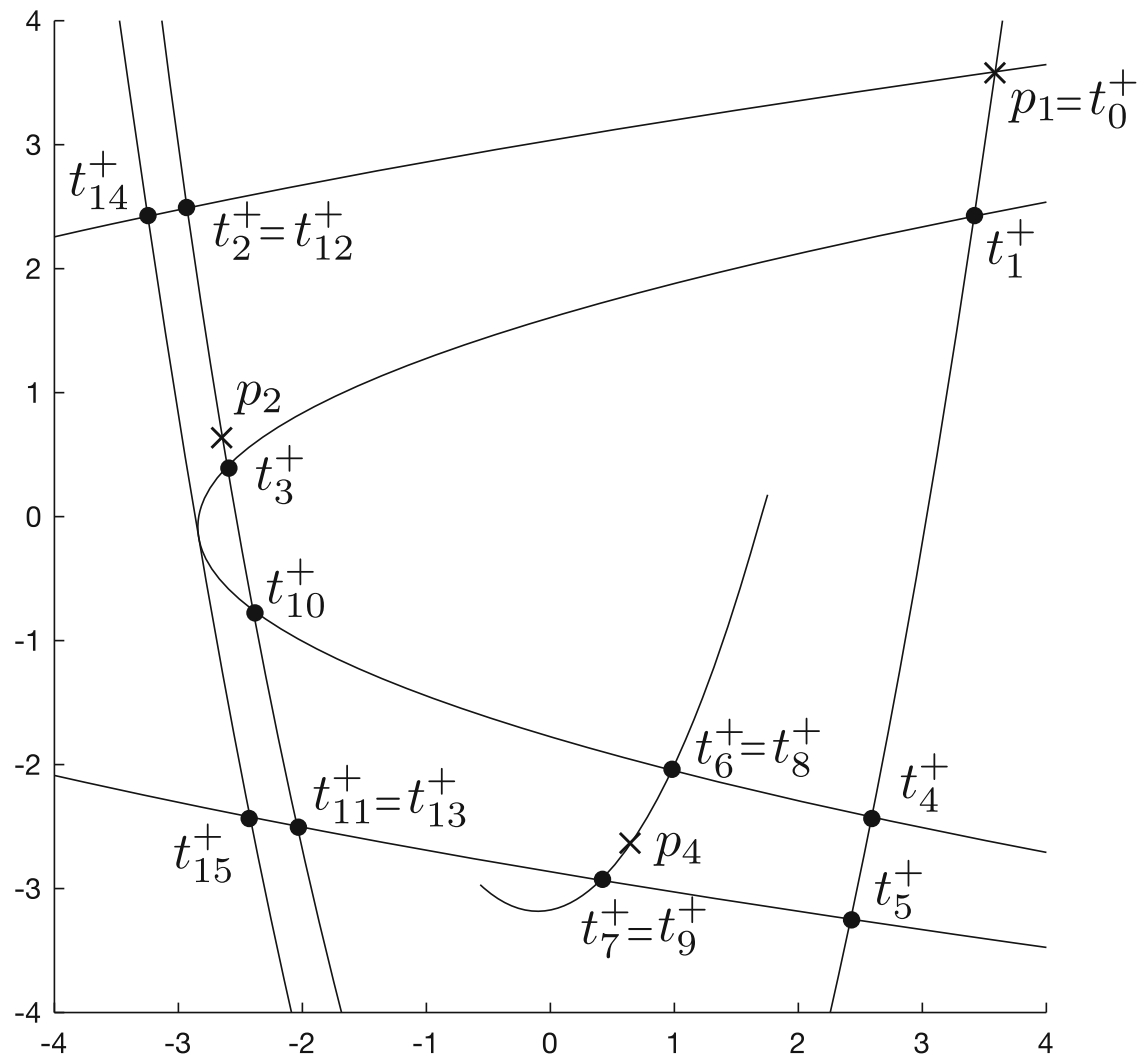
First Bifurcation Curve

[ZA-Ishii, ZA-Ishii-Takahasi] The boundary of the full horseshoe region of the Hénon map is given by a curve $b \rightarrow a_{\text{tgc}}(b)$ which is monotone and analytic on $b > 0$ and $b < 0$ with the following properties:

- (1) Hénon is hyperbolic full horseshoe iff $a > a_{\text{tgc}}(b)$
- (2) Top. entropy $< \log 2$ iff $a < a_{\text{tgc}}(b)$
- (3) When $a = a_{\text{tgc}}(b)$, there is a unique orbit of homoclinic tangency ($b > 0$) or heteroclinic tangency ($b < 0$)



Dynamics On the Curve



Theorem. Let $(a, b) \neq (a', b')$ be parameter values on the boundary of the full horseshoe region. Then $H_{a,b}$ and $H_{a',b'}$ are **NOT** topologically conjugate.

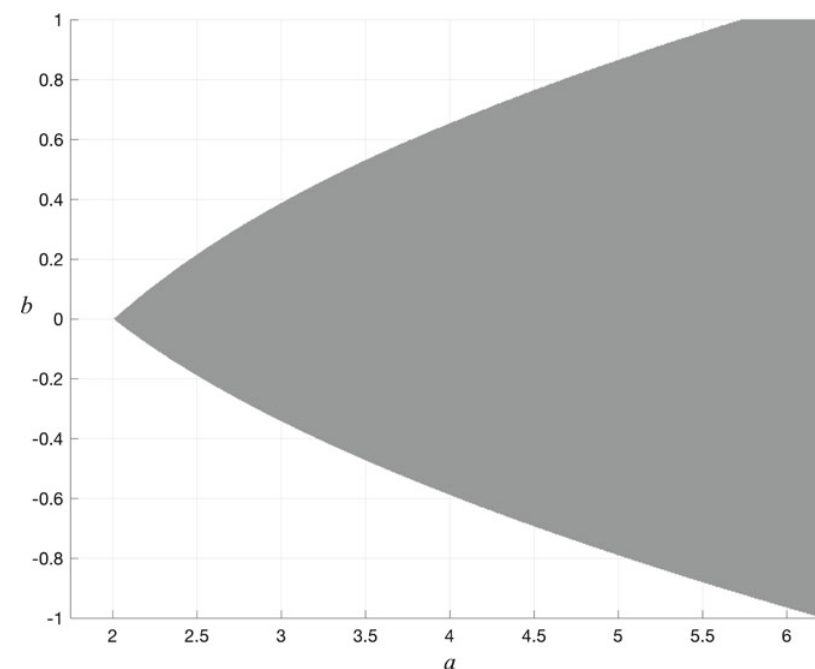
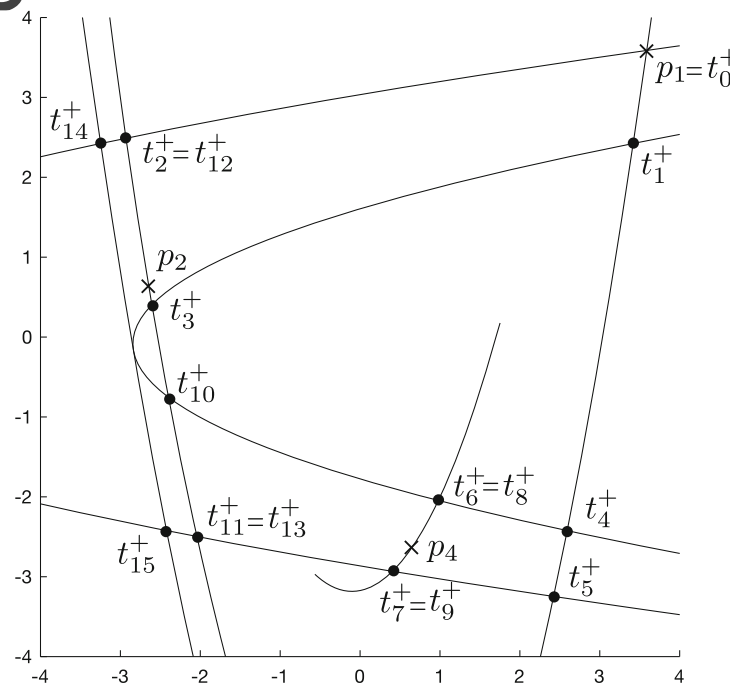
Remark 1

If (a, b) and (a', b') are on the first bif. curve,

- $H_{a,b}$ and $H_{a',b'}$ are not topologically conjugate

BUT,

- dynamics on their chain recurrent sets (= the filled Julia sets) are presumably topologically conjugate.



Remark 2

The description of the first bifurcation curve given in [ZA-Ishii, ZA-Ishii-Takahasi] heavily depends on the complex analytic nature of the Hénon map: we use

(1) Crossed Mapping Condition

- topological condition $\implies$ analytic conclusion

(2) Bedford-Smillie, Bedford-Lyubich-Smillie Theorems

- no tangency $\implies$ uniformly hyperbolic
- maximal entropy $\iff$
all homoclinic/heteroclinic intersections are real

Moduli of Stability

Moduli of Stability

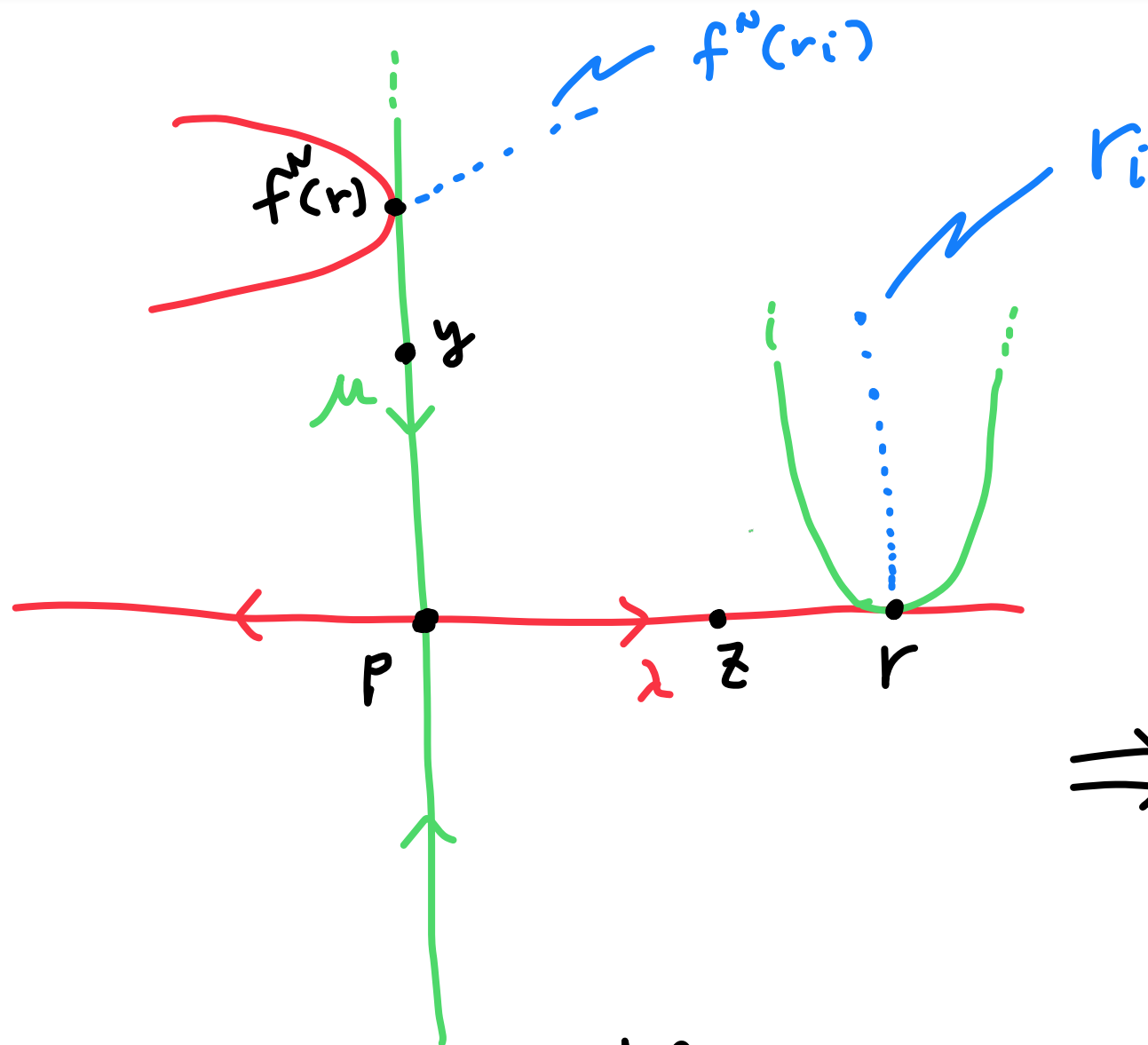
(homoclinic, 2dim version by R. A. Posthumus)

Let p (resp. p') be a hyp. saddle of f (resp. f') with a quadratic homoclinic tangency. Let μ and λ (resp. μ' and λ') be the contracting and expanding eigenvalue of the saddle. If there is a topological conjugacy between f and f' mapping p to p' , then we have

$$\frac{\log |\lambda|}{\log |\mu|} = \frac{\log |\lambda'|}{\log |\mu'|}$$

- Rem.**
- Original heteroclinic version is due to J. Palis.
 - In higher dim, we need a linearizing coordinate.

Why is it "topologically" invariant? (1/2)



$$r_i \rightarrow r$$

$$f^{-n_i}(r_i) \rightarrow y \in W_p^s$$

$$f^{m_i}(r_i) \rightarrow z \in W_p^u$$

$$\Rightarrow d(r_i, W_p^u) \approx C_1 \mu^{n_i}$$

$$d(r_i, W_p^s) \approx C_2 \lambda^{-m_i}$$

Since $d(r_i, W_p^u) \approx d(r_i, W_p^s)$

$$\frac{\log |\lambda|}{\log |\mu|} = - \lim_{i \rightarrow \infty} \frac{n_i}{m_i}$$

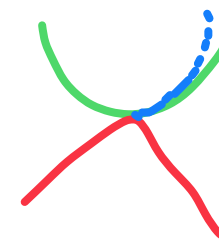
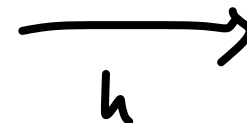
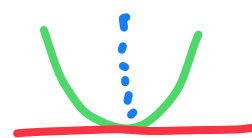
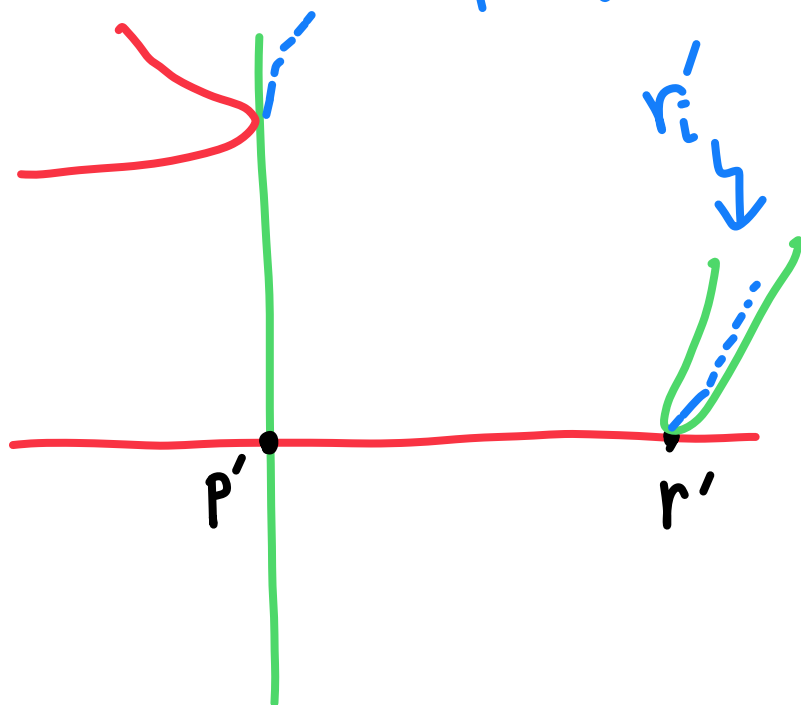
Why is it "topologically" invariant? (2/2)

$$\frac{\log |\lambda|}{\log |\mu|} = - \lim \frac{n_i}{m_i}$$

Now we apply a top. conj h : $r'_i := h(r_i)$

$$d(r'_i, w_{p'}^s) \not\approx d(r'_i, w_{p'}^u)$$

h is not smooth



However, we have

$$d(r'_i, w_{p'}^s) \leq d(r'_i, w_{p'}^u)$$

and $d(r'_i, w_{p'}^u) \approx c_1 \mu'^{n_i}$ $d(r'_i, w_{p'}^s) \approx c_2 \lambda^{-m_i}$

$$\Rightarrow \frac{\log |\lambda'|}{\log |\mu'|} \leq - \lim \frac{n_i}{m_i} = \frac{\log |\lambda|}{\log |\mu|}$$

Proof of the theorem

Sketch of the Proof (1/4)

We will show that

- A) the modulus is monotone increasing on the first bifurcation curve
- B) tangency is quadratic

Sketch of the proof ($0 < b \leq 1$)

- A) On the first bifurcation curve $b \rightarrow a_{\text{tgc}}(b)$, we have

$$\frac{d}{db} m(a_{\text{tgc}}(b), b) = \underbrace{\frac{dm}{da}}_{\text{positive}} \cdot \underbrace{\frac{da_{\text{tgc}}}{db}}_{\text{positive}} + \underbrace{\frac{dm}{db}}_{\text{positive}} > 0$$

Sketch of the Proof (2/4)

Let e^u and e^s be the unstable/stable eigenvalue of P , the fixed point with a homoclinic tangency.

$$e^u = \frac{1}{2} \left(\sqrt{4a + (b+1)^2} + b + 1 \right) + \sqrt{\frac{1}{4} \left(\sqrt{4a + (b+1)^2} + b + 1 \right)^2 - b}$$

$$e^s = \frac{1}{2} \left(\sqrt{4a + (b+1)^2} + b + 1 \right) - \sqrt{\frac{1}{4} \left(\sqrt{4a + (b+1)^2} + b + 1 \right)^2 - b}$$

$$\frac{de^u}{da} = \frac{\sqrt{4a + (b+1)^2} + b + 1}{2 \sqrt{4a + (b+1)^2} \sqrt{\frac{1}{4} \left(\sqrt{4a + (b+1)^2} + b + 1 \right)^2 - b}} + \frac{1}{\sqrt{4a + (b+1)^2}} > 0$$

$$\frac{de^u}{db} = \frac{1}{2} \left(\frac{b+1}{\sqrt{4a + (b+1)^2}} + 1 \right) + \frac{b \left(\sqrt{4a + (b+1)^2} + b + 2 \right) + 2a + 1}{2 \sqrt{4a + (b+1)^2} \sqrt{\frac{1}{4} \left(\sqrt{4a + (b+1)^2} + b + 1 \right)^2 - b}} > 0$$

Sketch of the Proof (3/4)

Since $e^u \cdot e^s = b$,

$$m(a, b) = \frac{\log e^s}{\log e^u} = \frac{\log b - \log e^u}{\log e^u} = \frac{\log b}{\log e^u} - 1.$$

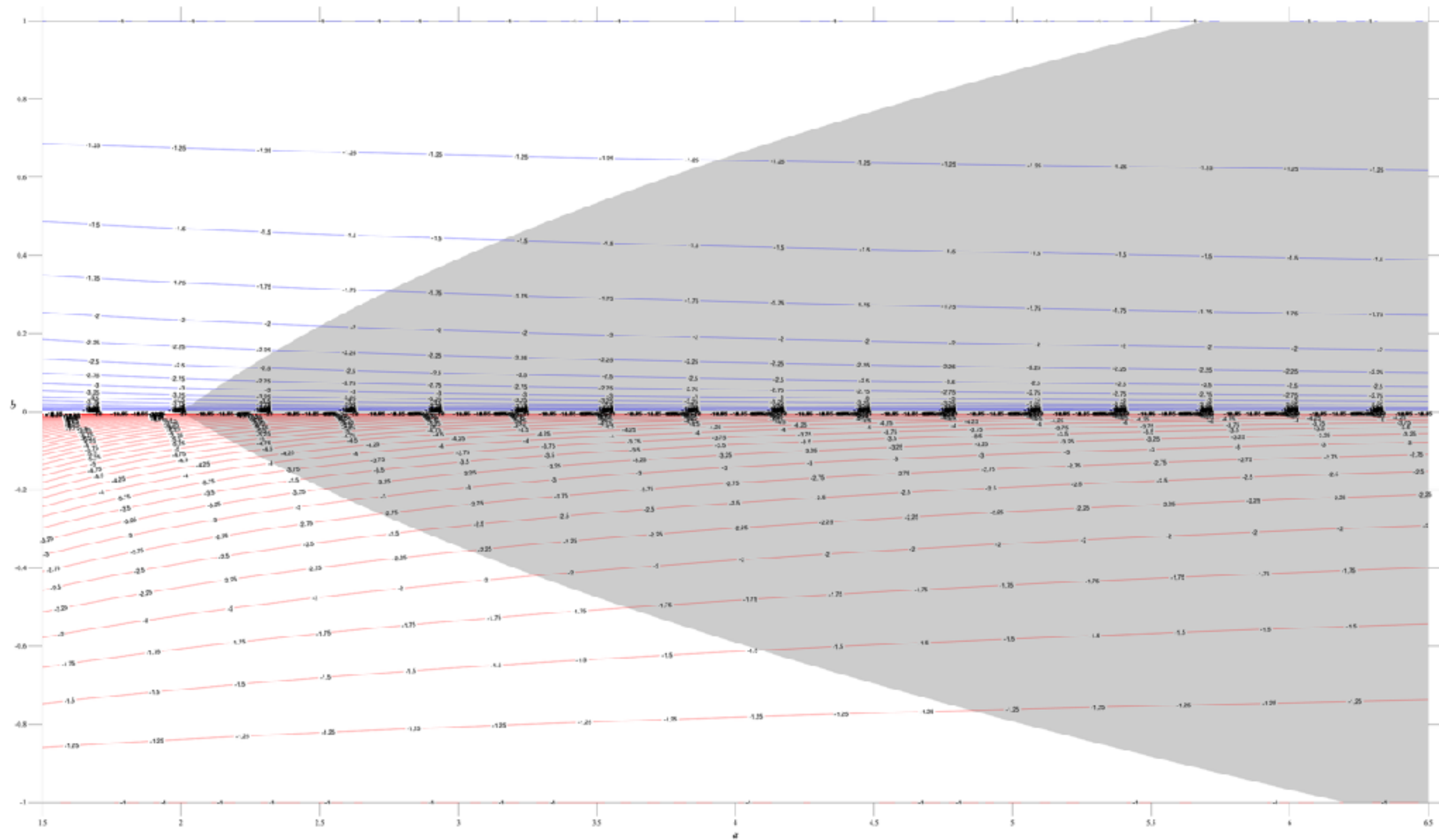
$$\frac{dm}{da} = \log b \cdot \frac{-\frac{d}{da} \log e^u}{(\log e^u)^2} = \frac{-\log b}{(\log e^u)^2} \cdot \frac{\frac{d}{da} e^u}{e^u} > 0$$

$$\frac{dm}{db} = \frac{1}{b} \frac{1}{\log e^u} + \log b \cdot \frac{-\frac{d}{db} \log e^u}{(\log e^u)^2} = \frac{1}{b} \frac{1}{\log e^u} + \frac{-\log b}{(\log e^u)^2} \cdot \frac{\frac{d}{db} e^u}{e^u} > 0$$

The case $-1 \leq b < 0$:

The proof goes similarly but involves some rigorous computer arithmetics.

Contour of the modulus



Sketch of the Proof (4/4)

B) To show that the tangency we take is quadratic, we use the following fact:

[Bedford–Smillie (2004) *Ann. Math.*] If the Hénon map has maximal entropy, but is not uniformly hyperbolic, then there exists a quadratic homoclinic/heteroclinic tangency with respect to a periodic point.

On the first bifurcation curve,

- entropy is maximal
- there exists a unique orbit of tangency

$\Rightarrow$ the tangency is quadratic

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